

Phishing Email Detection Model (Mini Project)

Objective:

Build a machine learning model using Scikit-learn to classify emails as **Phishing** or **Safe**.

Python Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, confusion_matrix

# Load dataset (CSV should contain 'text' and 'label' columns)
data = pd.read_csv("emails.csv")

X = data["text"]
y = data["label"]

vectorizer = TfidfVectorizer(stop_words="english")
X_vec = vectorizer.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_vec, y, test_size=0.2, random_state=42)

model = MultinomialNB()
model.fit(X_train, y_train)

predictions = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, predictions))
print("Confusion Matrix:")
print(confusion_matrix(y_test, predictions))

sample = ["Click here to verify your bank account immediately!"]
sample_vec = vectorizer.transform(sample)
result = model.predict(sample_vec)
print("Prediction:", result[0])
```

Key Features:

- Train on phishing and legitimate email dataset.
- Extract text features using TF-IDF.
- Classify emails as **Phishing** or **Safe**.
- Display accuracy and confusion matrix.

Expected Outcome:

The model classifies emails as Phishing or Safe with high accuracy based on textual content and extracted features.